

Good vibrations: Exploring how orbital resonance shapes the Galilean moons

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Upcoming missions to the Galilean moons will revolutionize our understanding of their interior evolution, largely governed by their orbital resonance. Yet, to fully understand the feedback between orbit and interior, crucial in determining their habitability potential, continued efforts are needed.

From discovery to first characterization

Since Galileo Galilei first resolved four faint points of light orbiting Jupiter in 1610, the Galilean moons have served as a natural laboratory for understanding how planetary systems work. With increasingly powerful ground-based observations, it became clear that Io and Europa are roughly the size of Earth's Moon, whereas Ganymede—the largest moon in the Solar System—is larger than Mercury, and Callisto's size is in between; see also Fig. 1. Before that, long-term observations already showed that the inner three moons, Io, Europa, and Ganymede, are locked in a 4:2:1 orbital resonance (the so-called Laplace resonance), which forces their orbits to remain slightly eccentric (i.e., non-circular).

When the Voyager 1 and 2 spacecraft visited Jupiter in 1979, the consequences of this eccentricity became apparent: the three inner moons exhibited intense activity. Io showed widespread volcanism so extreme that its internal power output exceeds that of Earth¹. Europa's icy surface was nearly devoid of impact craters, implying a young and continually renewed surface, while its fractured terrain hinted at a liquid ocean beneath the ice shell. Ganymede also displayed complex tectonic structures, hinting at a strong reheating event in the distant past that overturned large parts of its surface². In contrast, Callisto's heavily cratered surface indicated a great surface age and limited internal activity².

To investigate the system in greater detail, the Galileo spacecraft orbited Jupiter from 1995 to 2003. The estimate of the moons' moments of inertia (Mol), based on the measured degree-two gravity field, revealed a significant difference in internal mass distribution between the Galilean moons, and hence indicating a gradient in internal differentiation: the interiors of Io, Europa, and Ganymede are divided into distinct layers with different chemical compositions and densities, whereas Callisto shows a lower density contrast between layers³. Magnetic induction measurements provided evidence for subsurface oceans beneath Europa's ice shell and, surprisingly, also in the interior of Callisto⁴. For Ganymede, the Galileo mission revealed that this moon generates its own magnetic field in a (partially) liquid

metallic core³, similar to Mercury and the Earth, which hides the induction signal partly⁵. The presence of a subsurface ocean inside Ganymede was later confirmed from aurora fluctuations observed by the Hubble Space Telescope⁶. The discovery of liquid water beyond the snowline fundamentally reshaped our understanding of planetary habitability and expanded the range of potentially habitable environments.

The evolution of the inner three Galilean satellites and their subsurface oceans is governed by the strong tidal interactions between Jupiter (precisely characterized by the Juno mission) and its moons. The long-lived Laplace resonance, linking Io, Europa, and Ganymede, maintains a strong feedback between orbital dynamics and interior evolution. As Jupiter's gravity repeatedly stretches and compresses the moons along their eccentric orbits, internal friction generates heat, sustaining significant dissipation in Io and Europa over geological timescales. Differences in the magnitude and distribution of this tidal heating could explain the striking geological contrasts observed between the moons. Tidal heating also plays a fundamental role in the formation and long-term survival of their subsurface oceans, especially Europa's. The source of this heating is the moons' orbits, which are slowly evolving as a result⁷. Consequently, an understanding of the internal structures and evolution of these satellites is inseparable from the history of the Laplace resonance itself. Although there is a strong indication that Io has been tidally heated since formation⁸, the origin of the resonance remains debated. Whether the resonance is primordial or the result of a later capture implies markedly different thermal histories of the moons' hydrospheres^{9,10}. Characterizing the moons' habitability potential today thus tells us little about their longevity if we cannot reconstruct the orbital-thermal feedback loops that led to them.

Each moon's surface, interior, and dynamics encode signatures of different episodes of the resonance's history. The first step in characterizing these is to determine the density, thickness, and composition of the moons' internal layers (Fig. 1). Their internal differentiation is related to past and current heating, because establishing a density stratification requires high temperatures. Furthermore, the moons' tidal response (i.e., their deformation under Jupiter's gravity) is determined by their internal properties. The upcoming decade of exploration of the Galilean moons will provide invaluable constraints on the present state of the moons' orbits and interiors.

The 2030s: the era of detailed characterization

In the coming decade, three spacecraft will investigate Jupiter's icy moons, Europa, Ganymede, and Callisto. ESA's Jupiter Icy Moons Explorer (Juice), which will arrive in 2031, will have two close flybys of Europa, 20 of Callisto, and will ultimately orbit Ganymede¹¹. NASA's Europa Clipper will focus on Europa's habitability and, starting in 2030, will conduct at least 49 close flybys of Europa, 7 flybys of Ganymede,